25/7/26

Ex :- 2 Behaviour of Network Devices using Cisco Packet Tracer

Aim :-

To study and compare the Behaviour of a HUB and a switch using the cisco Packet Tracer simulator.

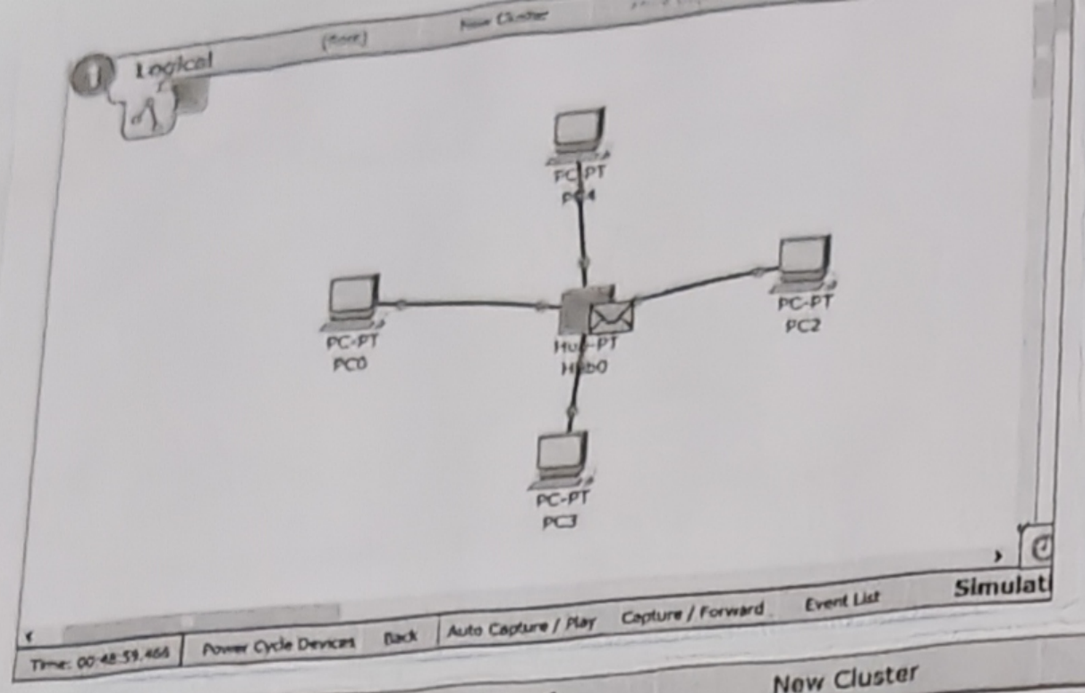
Procedure :-

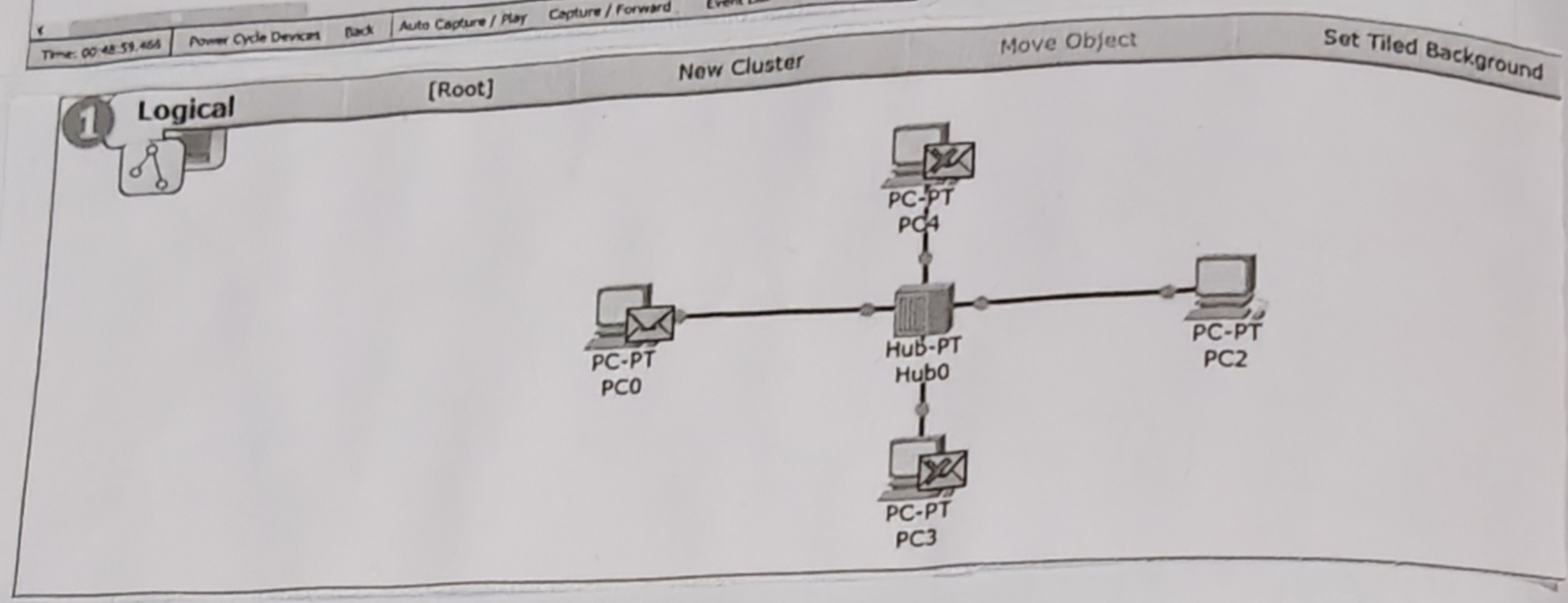
→ For HUB

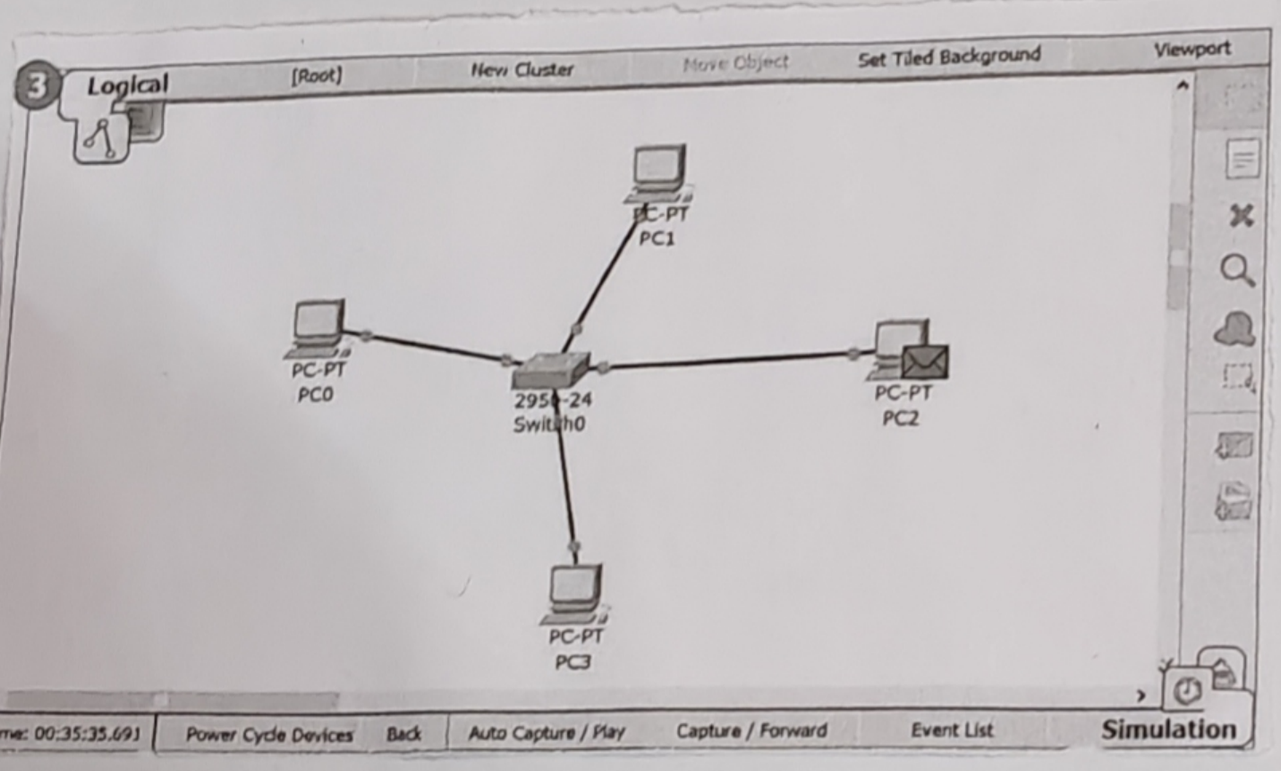
- Open cisco Packet Tracer
- Place 4 Generic PCs and 1 HUB in the workspace and connet all the PCs to the HUB using copper straight Through cables.
- Assign IP addresses and subnet masks to all the PCs. Click on simple PDU add.
- select one PC as source & another PC as destination.
- Run the simulation in realtime/ simulation mode & observe how PDU's forwarded through HUB.

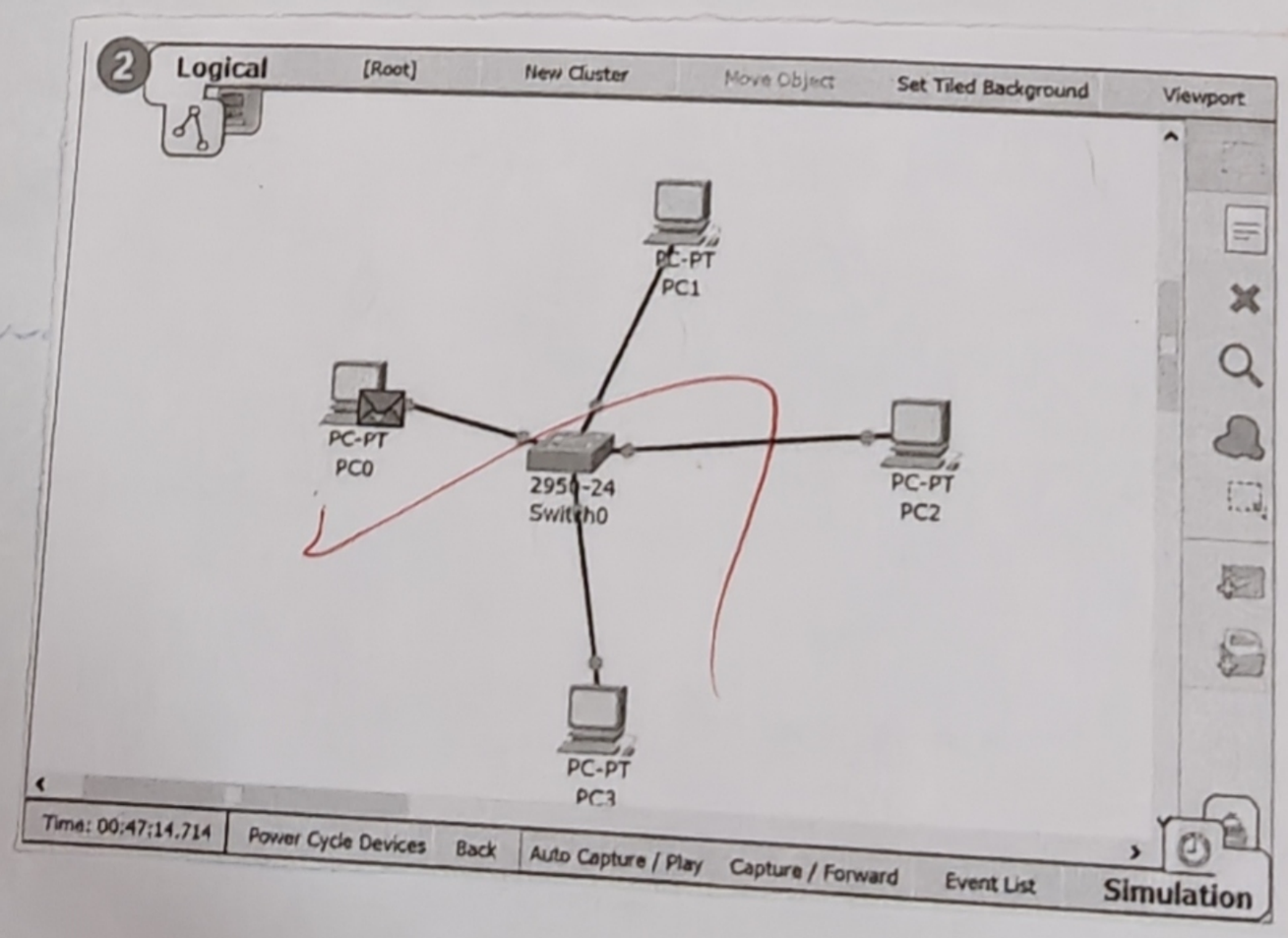
→ For Switch

- Place 4 Generic PCs and 1 switch
- connect all PCs to switch through cables


Logical
PC-PT
PC4
PC-PT
PC0
Hub-PT
Hub0
PC-PT
PC2
PC-PT
PC3
Time: 00:48:59.466
Power Cycle Devices
Back
Auto Capture / Play
Capture / Forward
Event List
Simulati


1
Logical
[Root]
New Cluster
Move Object
Set Tiled Background
PC-PT
PC4
PC-PT
PC0
Hub-PT
Hub0
PC-PT
PC2
PC-PT
PC3


3
Logical
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
PC-PT
PC1
PC-PT
PC0
2950-24
Switch0
PC-PT
PC2
PC-PT
PC3
Time: 00:35:35.691
Power Cycle Devices
Back
Auto Capture / Play
Capture / Forward
Event List
Simulation


2
Logical
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
PC-PT
PC1
PC-PT
PC0
2950-24
Switch0
PC-PT
PC2
PC-PT
PC3
Time: 00:47:14.714
Power Cycle Devices
Back
Auto Capture / Play
Capture / Forward
Event List
Simulation

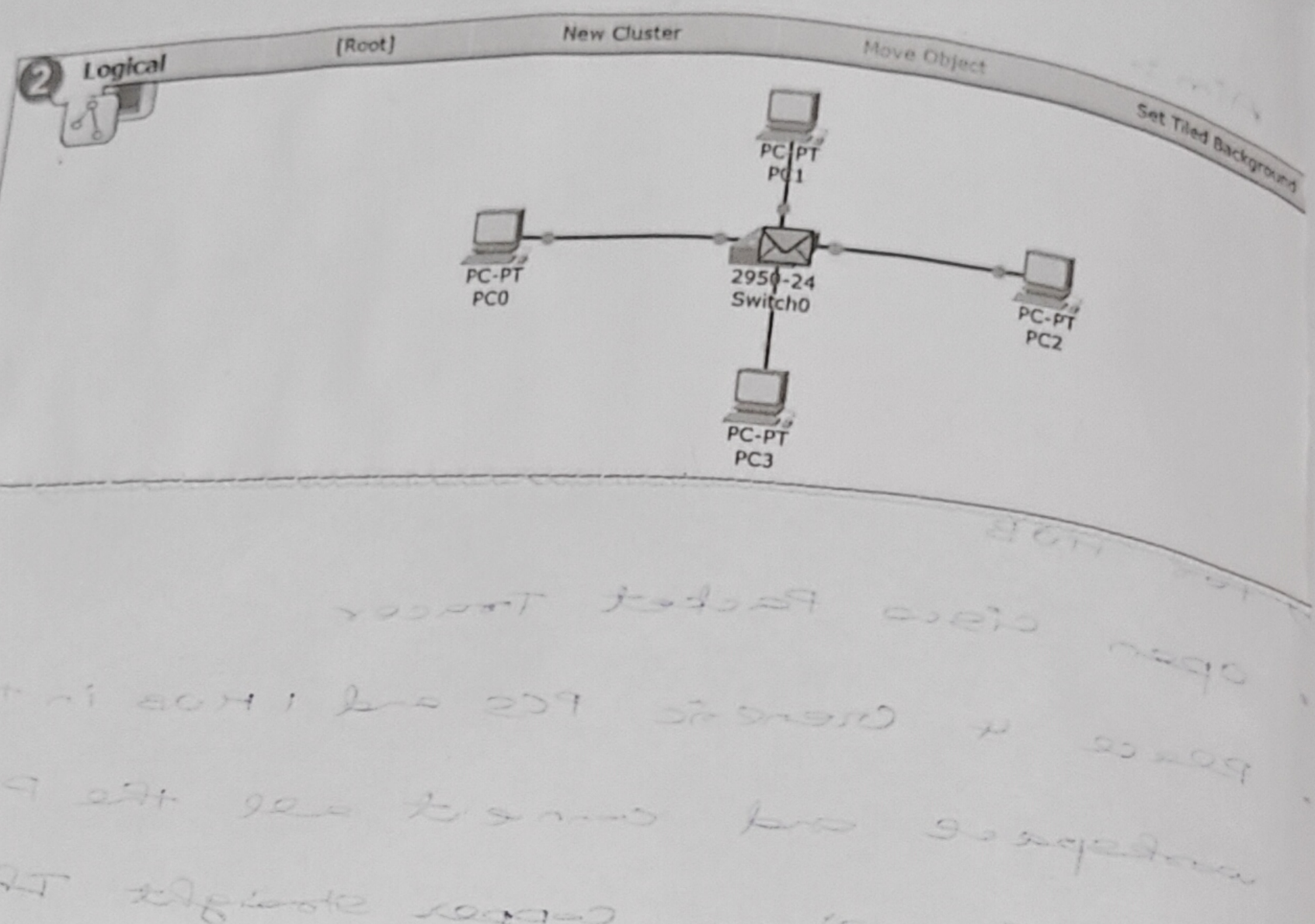
- Assign IP address and subnet masks
- click on Add Simple PDU.
- Select one PC as source & other as destination.
- Run the simulation in realtime.

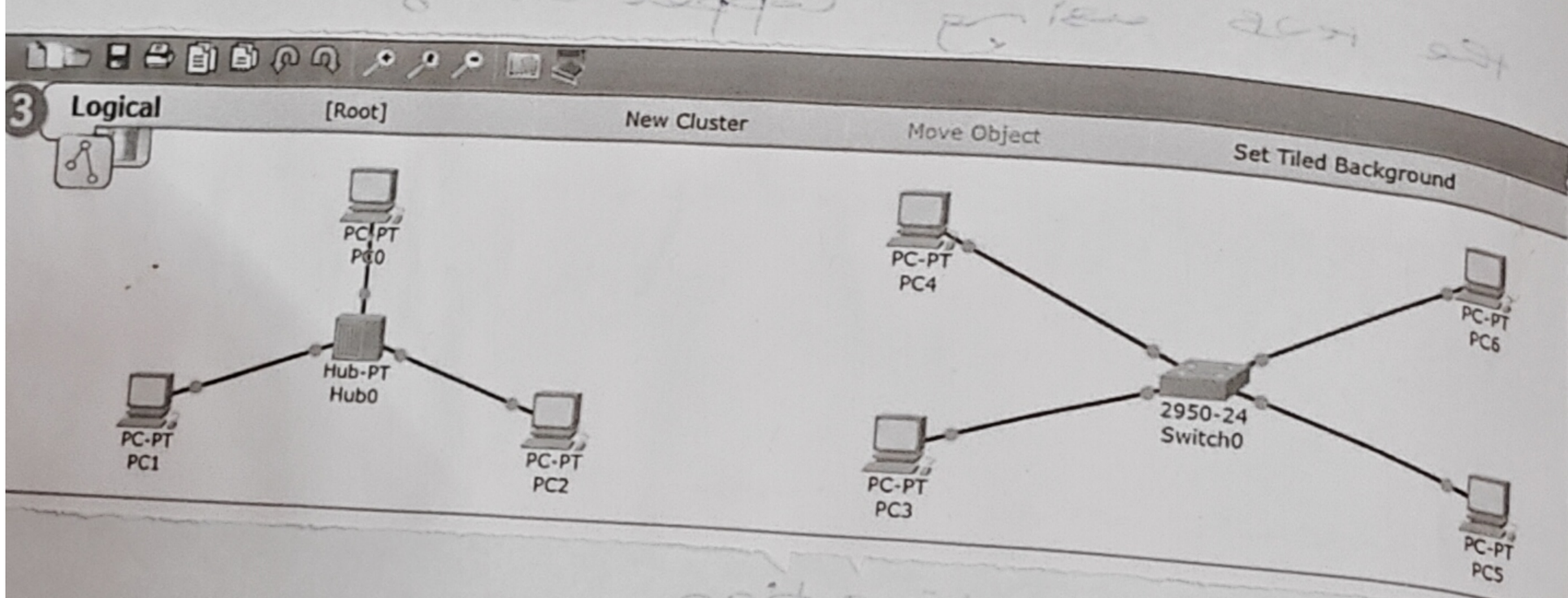
→ For combination of switch and Hub

- Place 8 generic PCs, 1 Hub and 1 switch
- Connect 4PCs to HUB using copper straight through cables & connect other 4PCs to switch.
- Assign IP address and subnet masks to all PCs from desktop → IP configuration option. click on Add Simple PDU
- Send PDU from one PC to another within HUB network and observe how HUB forwards packet.
- Next, send PDU from one PC to another within switch network.
- Run simulation in realtime/simulation mode and compare the behaviour of HUB and Switch.

Result :-

Behaviour of HUB and switch analysed successfully. It is observed that HUB Broadcasts data to all connected devices switch forwards data only to destination device.


2
Logical
(Root)
New Cluster
Move Object
Set Tiled Background
PC-PT
PC1
PC-PT
PC0
2950-24
Switch0
PC-PT
PC2
PC-PT
PC3


3
Logical
[Root]
New Cluster
Move Object
Set Tiled Background
PC-PT
PC0
Hub-PT
Hub0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC4
2950-24
Switch0
PC-PT
PC6
PC-PT
PC3
PC-PT
PC5